

NARWC Database Project — Progress Update

2026-07-19

NARWC Database Project

Progress Update

July 2026

Project Scope

- ▶ Migrate the historical NARWC sightings database from its legacy flat-file format into a modern, validated SQL database
- ▶ Build sustainable tools for curating and ingesting new survey data going forward
- ▶ Target: legacy migration complete and curation tools in usable shape by end of August 2026

What's Built: The Database

- ▶ A complete, modern SQL schema for the sightings database, rebuilt from the recovered legacy structure
- ▶ Covers all survey fields, lookup tables (species, behavior, platform, survey block, and more), and the relationships between them
- ▶ Ready to deploy

What's Built: The Migration Pipeline

- ▶ End-to-end pipeline: extract individual surveys from the legacy file, validate each one, upload to the database
- ▶ Uploads are transaction-safe — a failed upload rolls back cleanly rather than leaving partial data
- ▶ Runs against the full historical dataset (~1,200 surveys)

What's Built: Data Validation

- ▶ Every survey is checked against the rules that matter: required fields, coordinates, dates and times, species and behavior codes, environmental conditions, and cross-reference integrity
- ▶ Known, reviewed exceptions can be explicitly acknowledged and tracked — nothing is silently ignored or silently changed
- ▶ Curators can tune validation thresholds themselves (e.g. expected group sizes by species) without needing code changes

Where the Migration Stands

- ▶ The pipeline, schema, and validation are ready
- ▶ What remains is **data cleanup**, not tooling:
 - ▶ A known set of missing reference codes (species, behavior, platform, and similar lookup values) need confirmation
 - ▶ A known set of individual surveys have specific data issues that need review and correction
- ▶ Once those are resolved, the full historical dataset is ready to load

Still To Do

1. **Batch converters for the different input file formats** — each data provider currently sends surveys in its own format; these need dedicated parsers
2. **Test the git-based workflow** end to end
3. **Resolve the remaining data errors and warnings** in the historical data
4. **Get an updated data file from Bob** with the latest corrections
5. **Transfer the full system to NEAQ** — the big remaining milestone

Timeline

- ▶ Legacy migration: targeting completion before end of August 2026
- ▶ Curation tools: usable shape by end of August 2026
- ▶ Full system transfer to NEAQ: following migration completion